

InventOR: A Prompt-Configured, No-Fine-Tuning LLM Workflow for Material-Level Inventory Control

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Abstract

We present *InventOR*, a prompt-configured large-language-model (LLM) workflow that generates material-level (r, Q) inventory policies from cutoff-bounded historical CSVs and inline parameter blocks through an enterprise LLM application, without project-specific model training or fine-tuning. Deterministic parsing and post-cutoff simulation evaluate each LLM artifact against an SAP-derived and an SAP-safety-lead-time-informed operations-research comparator on a common 346-pair cohort from a three-plant industrial dataset. All 365 eligible plant-material pairs yield scoreable, capacity-feasible artifacts after retry handling a complete-cohort result that eliminates silent dropout. Working-day simulation reports 77.52% demand-weighted aggregate fill and 97.94% mean material fill for the LLM-emitted arm. Positive lost-sales penalty sensitivities remove the zero-penalty cost advantage. Stored metadata do not establish the exact deployed prompt or model/runtime identity; the contribution is an evaluation protocol with transparent boundary specification, manifest hashing, and prespecified exclusion rules whose first complete execution demonstrates complete-cohort

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reporting and the controls required for prospective evaluation.

Keywords: Generative Artificial Intelligence, Large Language Models, Operations Research, Supply Chain Management, Decision Support Systems, Explainable AI, Inventory Control

1. Introduction

Supply chain planning operates under volatile demand, uncertain replenishment, and disruption exposure [1]. Many material-control decisions still depend on simple parameters: reorder points, safety stock, order quantities, and review periods; these settings strongly affect service and inventory exposure [2, 3]. Classical operations research offers rigorous tools for stochastic and robust inventory decisions [4, 5, 6], but planners often need a narrower capability: converting imperfect historical records which are pervasive in practice [7] into transparent, defensible, auditable material-level policy values [8].

Large language models (LLMs) can help connect business intent, operational evidence, and analytical workflows, and autonomous decision-making is an emerging capability frontier for supply chain management [9, 10, 11]. Recommendations in inventory control require explainability, reproducibility, and meaningful human control before planners can act on them [12, 13]. This creates a design requirement for hybrid AI-supported OR: LLM artifacts must be inspectable, and numeric recommendations require explicit simulation and release controls [11, 13].

This paper addresses that requirement through *InventOR*, an operations-research-first workflow for material-level inventory decision support. An enterprise LLM application produces policy artifacts from material-scoped inputs; we parse and simulate their numeric values deterministically.

23 Our contribution is an evaluation protocol rather than a new inventory
24 model or solution algorithm. The comparator policies are standard textbook
25 constructions; what we add is a reusable, auditable procedure for deciding
26 whether LLM-emitted policy values are admissible and how they perform
27 against ERP-derived baselines on a common simulation horizon. Three com-
28 ponents make up that protocol.

29 **C1 Admissibility protocol for LLM-emitted policy artifacts.**

30 Material-specific inputs pass through a configured enterprise LLM
31 application without project-specific training or fine-tuning. De-
32 terministic scoreability and capacity checks decide admission; the
33 architecture reserves a planner release step, which we document as a
34 design requirement rather than an evaluated intervention [14].

35 **C2 Executable boundary specification.** We document the material-
36 input contract, alias-aware conversion, capacity validation, and simu-
37 lation boundary, so that a competing generator can be substituted and
38 scored under identical rules.

39 **C3 Complete-cohort reporting discipline.** All 365 eligible pairs from
40 233,078 parsed records yielded scoreable, capacity-feasible artifacts af-
41 ter retry handling; 346 pairs meet the common feasibility rules used
42 for the three-arm descriptive comparison. No pair is silently dropped,
43 and every exclusion rule is prespecified and reported.

44 The remainder of the paper is organised as follows. Section 2 reviews
45 related work and positions InventOR relative to recent AI-supported OR
46 contributions. Section 3 details the workflow, policy logic, and backtest
47 design. Section 4 reports completion and simulation results. Section 5 in-

48 interprets the findings and their limitations, and Section 6 summarises the
49 contribution and outlines future work.

50 **2. Literature Review**

51 Operations research has long addressed supply-chain uncertainty
52 through stochastic programming, robust optimization, and scenario-based
53 planning [4, 5, 6]. These methods formalize variability and recourse,
54 but operational inventory teams often face a more direct implementation
55 problem: maintaining reorder points, safety stock, safety lead time,
56 and order quantities from incomplete historical evidence in ERP-driven
57 planning environments [15, 16, 17, 18, 2, 3]. Static rules can be too
58 coarse, while sophisticated models can be difficult to explain and maintain.
59 This motivates decision support that keeps OR discipline while making
60 material-level parameter setting more transparent and auditable [8, 19].

61 Recent LLM research proposes natural-language interfaces, tool-using
62 agents, and pipelines that translate natural-language requests into optimiza-
63 tion model structures or executable code, including multi-agent systems that
64 make autonomous per-period inventory decisions or seek consensus on op-
65 erational actions [11, 20, 21, 22, 23, 24, 25, 26]. These approaches do not
66 specify how inventory histories should be reduced into demand regimes,
67 lead-time diagnostics, service buffers, and feasible policy parameters with
68 deterministic auditability.

69 We follow this hybrid view. We supply a filtered material CSV plus
70 an inline parameter block through an enterprise LLM application. The in-
71 tended prompt prohibits stochastic simulation, machine-learning libraries,
72 and unrestricted data fabrication, but stored run metadata do not inde-

pendently identify the deployed prompt or runtime. Because LLM outputs can contain fabricated or unfaithful content [27], our parser uses alias-aware lookups and nested-field resolution before requiring numeric positive reorder point and order quantity, numeric non-negative safety stock, and a resolvable simulator family. No project-specific model fitting or fine-tuning is performed. We treat the LLM as an artifact-generation component inside a bounded decision process. Table 1 positions our framework relative to recent AI-supported OR and inventory-control work.

Table 1: Literature positioning: InventOR relative to recent AI-supported OR contributions for inventory and supply chain decision support.

Study	LLM role	Policy class	Demand model	Analytical result	Evaluation
Huang et al. [20]	NL→code	General OR	Not fied	speci- None	Benchmark
Ahmaditeshnizi et al. [22]	NL→model	General OR	Not fied	speci- Solver-linked	Benchmark
Zhang et al. [23]	Reasoning agent	General OR	Not fied	speci- Model solve	and Case studies
Quan and Liu [24]	Multi-agent control	Inventory	Sequential demand	Per-period decisions	Simulation
Prak et al. [28]	None	Inventory control	Compound Poisson / intermittent	Robust estimation	Empirical study
Babai et al. [29]	None	Forecasting-linked inventory	Regime-specific	Review thesis	syn- Review
Wasserkrug et al. [11]	Interface	General	Not fied	speci- None	Conceptual
Cohen et al. [21]	Vision	General SCM	Not fied	speci- None	Vision statement
InventOR	Prototype	(r, Q) executed; (s, S) implemented, unexercised	Artifact-defined values	Descriptive	Post-cutoff simulation

The table highlights the paper’s narrower contribution: material-level parameter artifacts evaluated by a deterministic policy-family simulator. Unlike InvAgent’s sequential multi-agent decisions, InventOR emits one stationary policy per material from a bounded historical package; unlike Opti-

MUS and OR-LLM-Agent, it does not formulate or solve general optimization models. The (r, Q) framework is the workhorse of the continuous-review inventory literature [15, 18]; the intermittent-demand stream represented by Prak et al. [28] builds on Croston [30], Syntetos-Boylan [31], and Willemain et al. [32] to handle sporadic, non-normal demand patterns which our current prototype does not explicitly model.

3. Methodology

The unit of analysis is a plant-material pair. The framework records a chain of custody from historical records to planner-facing policy values by separating deterministic input preparation, configured LLM inference, schema parsing, deterministic simulation, and human review [8, 33]. Each LLM session receives one filtered, material-scoped historical CSV as a session resource and a small inline parameter block containing current cost, MOQ, service, SAP reference, and material-level storage-capacity fields. No project-specific model weights are trained, fine-tuned, or updated. Because the stored metadata omit prompt hash, application version, and model/runtime identifier, this study does not claim that the exact deployed instruction configuration is independently reproducible. Supplementary File S1 reproduces the intended application-side system and tool prompt texts verbatim, together with their disclosed discrepancies against the executable simulator.

3.1. Problem Definition: Single-Item Continuous-Review Control with Stress-Adjusted Service Buffering

The executed task is a single-item, single-echelon continuous-review simulation with lost sales [16, 17, 34]. Each plant-material pair is treated in-

110 dependently; multi-item coupling, shared capacity, substitution, and pro-
 111 duction scheduling are outside the pilot. For the SAP-SLT-informed OR
 112 comparator, pre-cutoff working-day demand yields mean demand $\bar{d}_i$, annu-
 113 alized demand $\bar{D}_i^{\text{ann}} = 250\bar{d}_i$, lead-time-demand dispersion, and stationary
 114 (R_i, Q_i) parameters. This comparator is not independent of ERP inputs be-
 115 cause source SAP safety lead time enters effective lead time. For the SAP-
 116 derived arm, source safety stock is retained while R_i and Q_i are recomputed
 117 by the same executable routines. For the LLM arm, scoreable policy pa-
 118 rameters originate directly in the artifact and pass alias-aware compatibility
 119 parsing plus strict executable-contract validation; the reference equations
 120 below do not reconstruct them. The simulator executes validated (r, Q) ar-
 121 tifacts as fixed-quantity policies and validated (s, S) artifacts as order-up-to
 122 policies. In this study all 365 scoreable artifacts resolved to the (r, Q) fam-
 123 ily after validation, so the order-up-to path is an available but unexercised
 124 capability rather than an evaluated result.

125 *Reference policy calculation.* The following equations describe the executed
 126 SAP-SLT-informed OR comparator; they do not reconstruct direct LLM-
 127 emitted policies. The order quantity $Q_i > 0$ is rounded upward to an integer,
 128 floored at source MOQ, and then projected to its material-level storage limit
 129 when possible. The reorder point $R_i \geq 0$ triggers replenishment. The EOQ
 130 anchor minimizes the standard annual cost approximation

$$\min_{Q_i} TC_i(Q_i; SS_i) = h_i \left(\frac{Q_i}{2} + SS_i \right) + K_i \frac{\bar{D}_i^{\text{ann}}}{Q_i},$$

131 where SS_i is arm-specific safety stock, $h_i = \text{holding_cost_rate}_i \cdot$
 132 unit_cost_i , and $K_i = \text{ordering_cost}_i$. The implementation first uses
 133 $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i(250\bar{d}_i)/h_i} \rceil\}$ when demand and ordering cost are

134 positive, and MOQ otherwise. For every arm, a supplied capacity requires
 135 $\text{MOQ}_i \leq Q_i$ and $SS_i + Q_i \leq \text{max_storage_units}_i$; comparator quantities
 136 above the residual limit are reduced, while pairs with $SS_i + \text{MOQ}_i$ above
 137 the limit are excluded from the common cohort. The SAP-SLT-informed
 138 OR safety stock is $SS_i = \lceil z_{\alpha_i} \sigma_{LTD,i} \rceil$, with $R_i = \lceil \bar{d}_i L_i^{\text{eff}} + SS_i \rceil$. The
 139 SAP-derived arm substitutes source SAP safety stock into the same reorder-
 140 point expression. No lot-size multiple or stress coefficient is executed. The
 141 primary holding/order cost has a zero shortage penalty; positive lost-sales
 142 penalties are sensitivity analyses.

143 *Core constraints.* Inventory evolves under a lost-sales continuous-review rule
 144 on the working-day decision set $\mathcal{W}$. All arms share the first observed post-
 145 cutoff on-hand inventory; the SAP-SLT-informed OR $R_i + Q_i$ is used only
 146 when that observation is unavailable. The initial on-order pipeline is empty.
 147 Duplicate same-material, same-date rows are aggregated by summing de-
 148 mand, receipts, corrections, and scrap and averaging positive lead times,
 149 but the simulator consumes only demand and excludes observed receipts,
 150 corrections, and scrap. The simulator rounds mean source lead time and
 151 schedules every modeled order by that number of subsequent working ob-
 152 servations:

$$I_{i,t} = \max(0, I_{i,t-1} + a_{i,t} - d_{i,t}), \quad t \in \mathcal{W},$$

153 where $a_{i,t}$ is the quantity whose working-day due observation is no later than
 154 t . Scheduled arrivals are added first, realized demand is served subject to
 155 lost sales, and the post-demand inventory position is computed as on-hand
 156 plus outstanding orders. For a validated (r, Q) artifact, a fixed quantity is

157 ordered when

$$\delta_{i,t} = \mathbf{1}\{\text{IP}_{i,t}^+ \leq R_i\}, \quad \text{IP}_{i,t}^+ = I_{i,t} + \sum_{s: \text{due}(s) > t} o_{i,s},$$

158 with $o_{i,t} = Q_i \delta_{i,t}$. For a validated (s, S) artifact, the same threshold test
 159 uses s_i , and an order of $S_i - \text{IP}_{i,t}^+$ is placed when positive; no artifact in
 160 this study triggered that branch. Orders due after the observed horizon
 161 remain in transit rather than being forced onto the final observation. Pa-
 162 rameters remain stationary. Shared observed initialization and the empty
 163 initial pipeline are part of the executed comparison design.

164 The reorder point targets a nominal cycle service level $\alpha_i \in [0.5, 0.999]$
 165 after numeric clamping. Under the normal approximation,

$$\mathbb{P}\left(D_{i,L_i^{\text{eff}}} \leq R_i\right) \geq \alpha_i \iff R_i \geq \bar{d}_i L_i^{\text{eff}} + z_{\alpha_i} \sigma_{LTD,i}.$$

166 The reported backtest metric is the ex-post fill rate, which is distinct from
 167 the nominal cycle-service target because it also depends on Q_i and realized
 168 demand [35].

169 Lead-time-demand variance is propagated under the demand and lead-
 170 time independence approximation,

$$\sigma_{LTD,i}^2 = L_i^{\text{eff}} \sigma_{D,i}^2 + \bar{d}_i^2 \sigma_{L,i}^2.$$

171 The executed SAP-SLT-informed OR buffer is

$$SS_i^{\text{OR}} = z_{\alpha_i} \sigma_{LTD,i},$$

172 with values rounded upward:

$$SS_i^{\text{OR}} = \lceil z_{\alpha_i} \sigma_{LTD,i} \rceil, \quad R_i = \lceil \bar{d}_i L_i^{\text{eff}} + SS_i^{\text{OR}} \rceil.$$

173 The executable statistics add non-negative source SAP safety-lead-time days
 174 to mean lead time when constructing comparator lead-time-demand disper-
 175 sion (Section 3.5 discusses this confound). LLM-emitted values remain ar-
 176 tifact values accepted by alias-aware parsing and strict executable-contract
 177 validation.

178 *Parser and review boundary.* Artifact policy labels determine simulator
 179 routing after validation. The parser resolves compatibility aliases, then
 180 requires a valid (r, Q) tuple or explicit s and S values for an (s, S) label;
 181 it rejects malformed, infeasible, or internally inconsistent artifacts. It does
 182 not execute label-specific optimization or deterministic escalation. Because
 183 no artifact supplied explicit and internally consistent s and S values, every
 184 scored policy in this study was executed as (r, Q) ; the reported evidence
 185 therefore concerns fixed-quantity control only. Planner review remains
 186 required before any operational use.

187 3.2. Historical Input Schema and Canonical Analysis Table

188 The uploaded CSV contains only the subset of source-table columns
 189 listed in Table 2. Plant and material identifiers are carried by the run key
 190 and inline block rather than repeated in the uploaded rows. The inline
 191 block passes cost, MOQ, service, SAP reference, and material-level storage-
 192 capacity variables. Stock-flow fields omitted from the uploaded subset are
 193 not analyzed by the executed LLM path.

Table 2: Fields included in the material-scoped CSV uploaded by the executed runner.

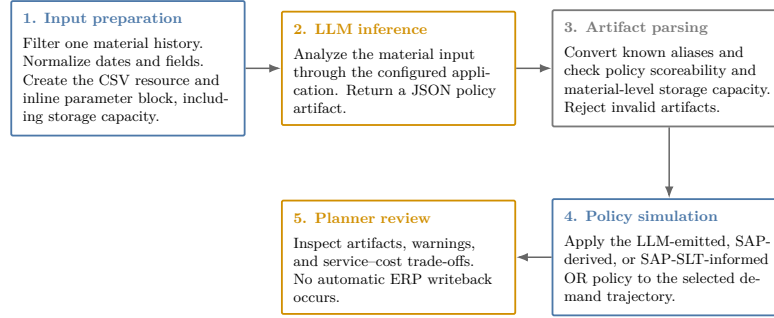
Column	Type	Description and method use
date	date / datetime	Calendar observation date; time ordering, aggregation, and diagnostics
actual_demand	numeric	Realized demand; mean, variability, intermittency, trend, and regime classification
delivery_time	numeric	Recorded replenishment lead time; rounded and scheduled as subsequent working observations by the simulator
minimum_order_quantity	numeric	Minimum replenishment quantity; order-size feasibility constraint
safety_stock_units	numeric	Existing safety stock; benchmarking against computed recommendations
safety_time_ workdays_actual	numeric	Source SAP safety-lead-time field; currently enters comparator effective lead time
supplier_plant_ distance	numeric	Supplier-to-plant distance; contextual lead-time-risk information
reliability_calc	numeric	Source supplier or lane reliability context
transportation_network	string	Source transportation-network context
service_level_target	numeric	Source service target used by comparator calculations
unit_cost	numeric	Purchase or standard cost per unit in the source dataset’s internal cost denomination; inventory-value proxy and input to holding-cost derivation and EOQ calculation
holding_cost_rate	numeric	Annual carrying-cost rate as a decimal (e.g., 0.20 for 20% of unit value); used to derive the annual cost of carrying one unit in stock, balance replenishment frequency against stock exposure, and compute economic order quantity
ordering_cost	numeric	Fixed replenishment cost in the source dataset’s internal cost denomination; input to EOQ calculation and simulated material-horizon holding/order cost
max_storage_units	numeric	Material-level upper bound applied to all evaluated arms: $MOQ \leq Q$ and $SS + Q \leq \text{max_storage_units}$
is_working_day	binary	Working-day flag; working-day demand statistics and non-zero demand share

194 3.3. Analytical Workflow

195 The workflow transforms a material snapshot into a policy artifact
196 through preparation, LLM inference, parsing, and simulation (Figure 1).

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: Prototype workflow from material input preparation to policy artifact and simulation output.

197 The runner uploads a normalized CSV and inline block; the application
 198 returns a JSON artifact. The parser converts it to an executable (r, Q)
 199 or (s, S) policy and rejects malformed outputs. Missing artifacts are re-
 200 tried. The simulator evaluates LLM, SAP-derived, and SAP-SLT-informed
 201 OR policies on the common horizon. An aggregate-only manifest hashes
 202 active-input, evaluation-code, and stored-artifact files; it records unavail-
 203 able provenance fields explicitly. Current runner code records these fields
 204 for future runs when the corresponding environment variables are supplied.
 205 The author declares retrospectively that evaluated artifacts were produced
 206 with an Anthropic Claude Sonnet 4.6 model behind the enterprise applica-
 207 tion, using a code-interpreter-style Python execution tool; this is recorded as
 208 unverified because no deployed model identifier or generation settings were
 209 persisted, and must not be read as machine-verified provenance. Stored
 210 inputs, artifacts, and code reproduce a simulation, not a fresh LLM comple-
 211 tion; planners retain implementation authority.

212 3.4. *Artifact Completeness and Future Readiness*

213 The completeness audit asks whether each eligible pair has a scoreable
214 artifact after retries; it does not estimate first-pass reliability, explanation
215 completeness, or deployment readiness. The empirical section is a retrospec-
216 tive audit of an already-executed workflow rather than a controlled exper-
217 iment: the generation cohort was produced before the evaluation protocol
218 was finalized, the deployed configuration was not frozen or instrumented in
219 advance, and no treatment was assigned. The post-cutoff simulation uses ar-
220 tifacts generated exclusively from pre-cutoff inputs and is descriptive rather
221 than a controlled prospective trial [36, 37]. A future evaluation must freeze
222 the deployed configuration, model/runtime identifiers, access controls, and
223 approval workflow.

224 3.5. *Safety Lead Time Computation*

225 Safety lead time (SLT) is an ancillary ERP-facing design field. The
226 prototype specification proposes combining transport, supplier, and quality
227 lead-time variability with a reliability-deficit premium [16, 3], but the present
228 release does not document estimators for those components. In the executed
229 comparator code, non-negative source SAP safety-lead-time days are added
230 to mean lead time before computing both SAP-derived and OR-computed
231 dispersion and reorder points (this coupling is also flagged in Section 3.1).
232 The OR arm is therefore labelled SAP-SLT-informed, and no separate SLT
233 effect is identified.

234 3.6. *Post-Cutoff Policy Simulation*

235 The confidential operational extract contains 233,078 parsed CSV
236 records, 411 plant-material pairs, and three anonymized plants (Plant

237 A, Plant B, and Plant C). The parameter cutoff is 2026-03-10 and the
 238 post-cutoff simulation horizon ends on 2026-08-27. Eligibility filters retain
 239 365 plant-material pairs with sufficient pre-cutoff and post-cutoff evidence:
 240 234 from Plant A, 65 from Plant B, and 66 from Plant C. Plant C has
 241 no positive source working-day flags, so its calendar flags are derived
 242 from same-date Plant A and Plant B votes; ties classify as working
 243 days. This deterministic repair permits a common decision calendar;
 244 because no alternative Plant C calendar exists, robustness is examined by
 245 reporting calendar-day arrivals and by re-running the full comparison on
 246 the native-calendar subset that excludes Plant C entirely. SAP-derived and
 247 SAP-SLT-informed OR parameters use pre-cutoff information. Stored LLM
 248 artifacts were generated from cutoff-bounded pre-cutoff inputs, so every
 249 arm is evaluated on the same post-cutoff horizon. Nineteen pairs cannot
 250 meet comparator safety stock plus MOQ within their supplied capacity;
 251 these are excluded before the common 346-pair comparison.

252 The simulated arms are: (i) SAP-derived, retaining source SAP safety
 253 stock while recomputing reorder point and EOQ order quantity from pre-
 254 cutoff statistics; (ii) SAP-SLT-informed OR, using the same statistics, source
 255 safety-lead-time input, and feasibility rules but without LLM inference; and
 256 (iii) LLM-emitted, a validated (r, Q) or (s, S) artifact. The simulator is
 257 continuous-review and lost-sales: all arms begin with the same first observed
 258 post-cutoff on-hand inventory, using the OR $ROP + Q$ only as a shared fall-
 259 back. The initial on-order pipeline is empty. On each working row, modeled
 260 arrivals due after rounded mean working-day lead time are received, demand
 261 is served subject to lost sales, post-demand inventory position is checked,
 262 and a policy-specific order is placed if its threshold is met. Observed post-
 263 cutoff receipts, corrections, and scrap are excluded. This common startup

design may create cutoff transients. Metrics are fill rate, aggregate demand-weighted fill rate, materials above 95% fill, zero-stockout materials, stockout days, average on-hand inventory, and simulated material-horizon cost.

For policy p and material i , the cost summaries use the simulated horizon of T_i working days. Let $\bar{I}_{i,p}$ be average on-hand inventory, h_i annual holding cost per unit, K_i fixed ordering cost, and $n_{i,p}^{\text{orders}}$ simulated orders. The implementation uses a constant 250 annual working days and computes

$$TC_{i,p} = h_i \bar{I}_{i,p} \frac{T_i}{250} + K_i n_{i,p}^{\text{orders}}.$$

This is a simulated material-horizon holding/order cost in the source dataset’s internal cost denomination, not an audited annual financial or booked accounting result. Table 3 reports arithmetic means across the cohort with zero shortage penalty. Sensitivities add lost-sales penalties of one and five times source unit cost; cost values must be read jointly with fill rate and stockout days.

4. Results

4.1. Completion and Artifact Validity

The final execution produced parser-scoreable artifacts for all 365 eligible plant-material pairs with zero missing and zero parser-rejected outputs after retry handling. Every artifact passes alias-aware parsing, nested-field resolution, and capacity validation against the supplied material-level storage bound; all resolve to the (r, Q) family after validation.

4.2. Main Three-Arm Exploratory Simulation

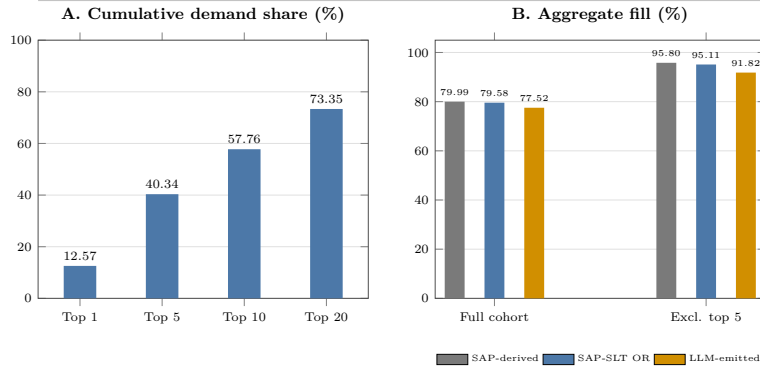
Table 3 reports three policy arms on the same 346 common-feasible material trajectories under working-day post-cutoff simulation. All arms satisfy identical MOQ and supplied per-material storage rules; 19 otherwise eligible pairs whose comparator safety stock plus MOQ exceed capacity are excluded.

Table 3: Post-cutoff working-day simulation on 346 common-feasible plant-material pairs. All arms use pre-cutoff parameterization or cutoff-bounded inputs and satisfy identical MOQ and supplied storage rules. Cost is mean simulated material-horizon holding/order cost with zero shortage penalty. SO days is the total count of stockout days across the cohort.

Policy	<i>n</i>	Mean FR (%)	Agg. FR (%)	$\geq 95\%$ (%)	Zero SO (%)	Mean cost	SO days
SAP-derived	346	98.38	79.99	94.51	80.35	3,184.20	741
SAP-SLT OR	346	98.27	79.58	93.93	88.73	3,852.22	799
LLM-emitted	346	97.94	77.52	92.49	81.50	2,848.96	936

Demand concentration and fill sensitivity

Largest-demand materials strongly influence full-cohort aggregate fill.



Generated from the working-days primary backtest; common-feasible cohort $n=346$.

Figure 2: Demand concentration and aggregate-fill sensitivity. Panel A reports cumulative demand share for the largest materials. Panel B compares full-cohort aggregate fill with the value after excluding the five largest-demand materials.

Relative to SAP-derived, LLM-emitted policies show 2.48 percentage points lower aggregate fill, 335.24 lower mean holding/order cost, and 195

292 additional stockout days. Relative to SAP-SLT-informed OR, 2.07 percent-
293 age points lower aggregate fill, 1,003.26 lower mean holding/order cost, and
294 137 additional stockout days.

295 Aggregate evidence is demand-concentrated. The largest 1, 5, 10, and
296 20 materials contribute 12.57%, 40.34%, 57.76%, and 73.35% of post-cutoff
297 demand. Excluding the five largest-demand materials raises aggregate fill
298 to 91.82% for LLM-emitted, 95.80% for SAP-derived, and 95.11% for SAP-
299 SLT-informed OR, versus 77.52–79.99% in the full cohort (Figure 2). The
300 largest-demand items drive aggregate fill.

301 At material level, LLM-emitted fill is not lower than SAP-derived for
302 314 of 346 materials (90.8%), while its cost is no higher for 238 (68.8%);
303 both conditions hold for 211 (61.0%). Against SAP-SLT-informed OR,
304 the corresponding counts are 309 (89.3%), 303 (87.6%), and 269 (77.7%).
305 Paired LLM-minus-comparator fill differences have first quartile, median,
306 and third quartile of 0.00 percentage points for both comparators, reflecting
307 many exact fill ties. Corresponding cost differences are -259.24 , -47.17 ,
308 and 34.96 versus SAP-derived and -787.83 , -209.32 , and -25.26 versus
309 SAP-SLT-informed OR. A deterministic 10,000-resample paired bootstrap
310 gives LLM-minus-SAP fill and cost means of -0.44 percentage points (95%
311 interval $[-0.99, -0.02]$) and -335.24 (95% interval $[-629.16, -105.29]$);
312 against SAP-SLT-informed OR, the corresponding values are -0.33 per-
313 centage points ($[-0.89, 0.14]$) and $-1,003.26$ ($[-1,363.81, -705.03]$). These
314 descriptive resampling intervals and aggregate results show service–cost
315 trade-offs rather than uniform policy ranking.

316 The artifact audit covers all 365 scoreable policies: normalized policy-
317 family agreement is 90.68%, and 42 artifacts meet a prespecified diagnostic-
318 outlier rule (CV absolute error above 0.75 or LLM-to-pipeline safety-stock

ratio outside $[0.5, 2.0]$). Removing the 23 flagged artifacts present in the common cohort raises LLM aggregate fill from 77.52% to 79.55%, while SAP-derived reaches 80.08% and SAP-SLT-informed OR reaches 79.70%; this post-hoc diagnostic subset does not establish a causal correction. Replacing primary working-day receipt offsets with calendar-day offsets gives aggregate fills of 77.77%, 80.25%, and 79.84%, respectively. Removing the one plant whose source working-day flags are entirely absent, and whose calendar is therefore imputed, leaves 280 native-calendar pairs with aggregate fills of 74.03% for LLM-emitted, 76.55% for SAP-derived, and 76.35% for SAP-SLT-informed OR, mean material-level fills of 98.19%, 98.71%, and 98.54%, zero-stockout shares of 81.79%, 88.21%, and 90.71%, and mean holding/order costs of 2,893.68, 3,471.66, and 4,168.76. In this native-calendar subset both comparators retain higher fill and higher zero-stockout shares than the LLM arm, so the imputed calendar does not manufacture the observed service gap. With a lost-sales penalty of one times unit cost, mean total costs are 1,688,922.75, 1,631,321.28, and 1,692,746.91, respectively; at five times unit cost, they are 8,433,217.92, 8,143,869.56, and 8,448,325.64. Thus, zero-penalty holding/order cost cannot establish economic superiority. These sensitivities do not resolve empty-pipeline, excluded-flow, or calendar-repair limitations.

5. Discussion

We identify three findings as the principal scientific contribution. First, 100% of 365 eligible plant-material pairs yield scoreable, capacity-feasible policy artifacts after retry handling a complete-cohort result that eliminates silent dropout and establishes a benchmark for future LLM-based inventory

344 studies. Second, all scored artifacts resolve to the (r, Q) family; the LLM
345 constructs every policy value from only the material-scoped history and
346 parameter block, while both comparators are warm-started from ERP set-
347 tings that embed accumulated planner judgment. The observed aggregate-
348 fill deficit of roughly two percentage points, and the higher stockout-day
349 count (936 versus 741 for SAP-derived and 799 for SAP-SLT-informed OR),
350 are therefore cold-start characteristics of a generator that receives no safety-
351 stock legacy or safety-lead-time input. An ERP-independent cold-start OR
352 arm is the comparator required to isolate generator quality and is specified
353 as a future-work requirement. Third, positive lost-sales penalty sensitivi-
354 ties remove the zero-penalty cost ranking, confirming that the apparent cost
355 advantage is driven by the penalty assumption rather than by generator
356 quality.

357 The evaluation protocol is the reusable contribution. An aggregate-only
358 manifest hashes active inputs, evaluation code, and stored artifacts, flagging
359 unavailable provenance fields. The input-contract specification, alias-aware
360 parser, capacity-validation rules, and the full exclusion log form a repro-
361 ducible boundary within which any LLM or OR generator can be substi-
362 tuted and scored under identical rules. The tracked historical 315-material
363 snapshots are byte-identical rather than distinct generation replicates [38];
364 repeat inference at cohort scale with fixed instrumentation belongs among
365 future-work requirements.

366 Our study is bounded by a single anonymized three-plant dataset, one
367 generation configuration, and a post-cutoff simulation that excludes ob-
368 served receipts, corrections, and scrap with an empty initial pipeline. These
369 are explicit design choices, not undisclosed weaknesses. The contribution is
370 the protocol and its first complete execution, not a performance claim.

371 6. Conclusion

372 We present *InventOR*, an applied AI/OR prototype that treats LLMs as
373 candidate-policy generators within deterministic preparation, parsing, and
374 simulation. The framework preserves an inspectable link between mate-
375 rial histories, policy fields, and simulated outcomes without project-specific
376 model training or fine-tuning.

377 The executed artifact set contains 365 scoreable policies generated from
378 cutoff-bounded, pre-cutoff material histories, all satisfying their material-
379 level storage bounds. On the 346-pair cohort where all arms satisfy identi-
380 cal MOQ and supplied storage rules, LLM-emitted aggregate fill is 77.52%,
381 mean material-level fill is 97.94%, mean simulated material-horizon hold-
382 ing/order cost is 2,848.96, and stockout days total 936 under working-day
383 receipt scheduling. The warm-start asymmetry explains the fill deficit and
384 higher stockout-day count relative to ERP-informed comparators. Positive
385 shortage-cost sensitivities remove the zero-penalty cost advantage. These
386 values are descriptive simulated-comparison evidence, not deployment or
387 production-trial evidence.

388 Future work should supply an ERP-independent cold-start OR arm, re-
389 peat inference at cohort scale to estimate generation variance, extend the ad-
390 missible policy family beyond (r, Q) to order-up-to and intermittent-demand
391 regimes, and run a prospective evaluation with planner-in-the-loop review
392 and fixed instrumentation. An ex-ante diagnostic-outlier exclusion rule and
393 a documented SLT-component estimator belong in the same evaluation pro-
394 tocol upgrade.

395 **Author Contributions**

396 Conceptualization, Methodology, Software, Formal analysis, Validation,
397 Visualization, Writing – original draft, Writing – review and editing: Aris
398 Dressino.

399 **Funding**

400 This research did not receive any specific grant from funding agencies in
401 the public, commercial, or not-for-profit sectors.

402 **Declaration of Competing Interest**

403 The author declares no known competing financial interests or personal
404 relationships that could have appeared to influence the work reported in this
405 paper.

406 **Data and Code Availability**

407 The raw operational data and material-level artifacts cannot be publicly
408 released because they contain commercially sensitive material, supplier, and
409 planning information. A confidentiality-screened release package contain-
410 ing analysis code, the anonymized variable schema, aggregate evaluation
411 results, manuscript sources, and aggregate integrity manifest is available
412 at github.com/DressPD/inventor_public. The release package is addi-
413 tionally deposited in a Zenodo archive whose version-specific DOI will be
414 registered at acceptance and cited in the final version of record; the GitHub
415 repository is the development mirror and the Zenodo deposit is the citable

416 archival copy. Supplementary File S1 accompanies this submission and con-
417 tains the intended application-side prompt specification. The historical de-
418 ployed application configuration, model/runtime identifier, inference dates,
419 and prompt hashes were not preserved and are not claimed as available; the
420 current runner records these fields for future executions.

421 **7. Acknowledgments**

422 The author thanks the industrial collaborators and material-control ex-
423 perts who contributed domain knowledge and access to anonymized oper-
424 ational data. The author also acknowledges the associated enterprise data
425 and IT teams for supporting data access and prototype execution. No for-
426 mal full-population planner evaluation is claimed. Any errors or omissions
427 remain the responsibility of the author.

428 **Declaration of Generative AI and AI-Assisted Technologies in the** 429 **Writing Process**

430 During manuscript preparation, the author used OpenAI GPT-5.6
431 through OpenCode for language editing, consistency checks, citation
432 review, and LaTeX proofing support; this is a writing-assistance tool and
433 is distinct from the Anthropic Claude Sonnet 4.6 deployment evaluated in
434 the research workflow. The author reviewed and edited the content after
435 use and takes full responsibility for the content of the published article.
436 Figures were rendered deterministically from author-reviewed LaTeX and
437 Python source; generative AI did not directly create or manipulate image
438 content. The research workflow evaluated in the paper is itself the object
439 of study and is described separately in the methodology.

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